

Problem framing, goal association, assumptions and associated research papers

|             |                      |          |                              |
|-------------|----------------------|----------|------------------------------|
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**Research Question**

Which cycloidal speed reducer architecture will fulfill NT-N-G1 and NT-N-G7?

**Abstract**

This document collates and analyses multiple research papers and notes for fulfilling Node v0 design goals in relation to cycloidal speed reducer design. Its intended purpose is to isolate observations, assumptions, risks, and design direction before committing to a detailed technical design.

**Keywords:** integrated actuator, mechanical design, cycloidal speed reducer, robotics, research notes

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## 1 RESEARCH SNAPSHOT

### Research Snapshot

|                         |                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem</b>          | Build a cycloidal speed reducer to maximise torque density and minimise backlash, making it easy to integrate with existing motor housing architectures and reducing weight. |
| <b>Key Finding</b>      | TBD                                                                                                                                                                          |
| <b>Main Uncertainty</b> | TBD                                                                                                                                                                          |
| <b>Next Step</b>        | Design the first iteration of the cycloidal speed reducer, and do a technical writeup of the design decisions and their impact on the requirements.                          |

## 2 PROBLEM FRAMING

### Research Question – System Direction

What cycloidal speed reducer architecture achieves the QDD required 9:1 gear reduction ratio while maximising efficiency (W/W %), minimising backlash (arcmin) and ensuring durable and long-lasting operation (cycles/failure) under the specified load and speed conditions?

## 3 PRELIMINARY DIRECTION

This research document will collate results from multiple research papers and notes to inform the design of a cycloidal speed reducer for Node v0. The goal is to identify the most promising cycloidal speed reducer architecture that meets the specified design goals (NT-N-G1, NT-N-G7). The motor housing architecture has been designed and the first iteration will inform the cycloidal speed reducer design. The motor housing output interface is the only interface that the cycloidal speed reducer will need to integrate with, and therefore the cycloidal speed reducer design will not be overly constrained by the motor housing architecture.

## 4 ASSUMPTIONS

- Cycloidal speed reducers are the most suitable architecture for Node v0, due to their high torque density, low backlash, and compact form factor.
- Steel is the most suitable material for the cycloidal disks and roller pins due to its high strength and wear resistance, and aluminium is suitable for the other components, due to its machinability.

## 5 CYCLOIDAL SPEED REDUCER OPERATING PRINCIPLES

Cycloidal speed reducers are named thus because of the profile of the internal cycloid plate gear, which matches the profile of a cycloidal curve. A cycloidal curve is created by tracing a point on the circumference of a circle, as it rolls without slipping along another circle. This is demonstrated in the following figure:



**Figure 1:** Cycloidal curve generation

The key advantage of a cycloidal reducer is that the plate gear meshes at all points with the lobes, guaranteeing no backlash in a perfectly manufactured reducer, and the rolling contact between the plate gear and the lobes reduces wear and increases efficiency.

### 5.1 Cycloidal Reducer Architectures

There are 4 different types of cycloidal speed reducer architectures:

- The stationary ring gear type epicycloid reducer
- The rotating ring gear type epicycloid reducer
- The stationary ring gear type hypocycloid reducer
- The rotating ring gear type hypocycloid reducer

## REFERENCES

- [1] Joong-Ho Shin, Soon-Man Kwon, On the lobe profile design in a cycloid reducer using instant velocity center, Mechanism and Machine Theory, Volume 41, Issue 5, 2006, Pages 596-616, ISSN 0094-114X, <https://doi.org/10.1016/j.mechmachtheory.2005.08.001>.